

Summary - Section 1

Mass flow rate

$$\dot{m} = \iint_A \rho \vec{V} \cdot \hat{n} \, dA$$

Conservation of mass/ continuity:

Integral or global form:

$$\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \, d(\text{vol}) + \oiint_A \rho \vec{V} \cdot \hat{n} \, dA = 0$$

Differential or local form:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$

Summary - Section 1

Conservation of momentum:

Integral or global form:

$$\begin{aligned} \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \vec{V} \, d(\text{vol}) = & - \iint_A \rho \vec{V} (\vec{V} \cdot \hat{n}) \, dA - \iint_A p \hat{n} \cdot dA \\ & + \vec{F}_{visc} + \vec{F}_{ext} + \iiint_{\text{vol}} \rho \vec{f}_b \, d(\text{vol}) \end{aligned}$$

Differential or local form:

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla p + \vec{f} + \frac{1}{\rho} (\vec{F}_{visc} + \vec{F}_{ext})$$

Summary - Section 1

Conservation of energy:

Integral or global form:

$$\begin{aligned} \frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) d(\text{vol}) = \\ - \oint_A \rho \left(u_e + \frac{1}{2} \rho \vec{V} \cdot \vec{V} + gz \right) \vec{V} \cdot \hat{n} dA \\ + \dot{Q} - \oint_A p \vec{V} \cdot \hat{n} dA + \dot{W}_{visc} + \dot{W}_s + \iiint_{\text{vol}} \rho \vec{f}_b \cdot \vec{V} d(\text{vol}) \end{aligned}$$

Differential or local form:

$$\begin{aligned} \frac{\partial}{\partial t} \rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) + \nabla \cdot \left[\rho \left(u_e + \frac{1}{2} \vec{V} \cdot \vec{V} + gz \right) \vec{V} \right] = \\ \dot{q} - \nabla \cdot (p \vec{V}) + \dot{w}_{visc} + \rho \vec{f}_b \cdot \vec{V} \end{aligned}$$

Summary - Section 2

Eqn. of a streamline:

$$w \, dy - v \, dz = u \, dz - w \, dx = v \, dx - u \, dy = 0$$

Angular velocity:

$$\vec{\omega} = \frac{1}{2} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \frac{1}{2r} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ v_r & rv_\theta & v_z \end{vmatrix} = \frac{1}{2} \vec{\nabla} \times \vec{V}$$

Circulation:

$$\Gamma = - \oint_C \vec{V} \cdot d\vec{s} = - \iint_A (\vec{\nabla} \times \vec{V}) \cdot \hat{n} dA$$

Summary - Section 2

Velocity potential:

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}, \quad w = \frac{\partial \phi}{\partial z}$$
$$v_r = \frac{\partial \phi}{\partial r}, \quad v_\theta = \frac{1}{r} \frac{\partial \phi}{\partial \theta}$$

Stream function:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$
$$v_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta}, \quad v_\theta = -\frac{\partial \psi}{\partial r}$$

Summary - Section 2

Laplace equation:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}$$

Divergence:

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \frac{1}{r} \frac{\partial (rv_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z}$$

Curl:

$$\vec{\nabla} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix} = \frac{1}{r} \begin{vmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ v_r & rv_\theta & v_z \end{vmatrix}$$

Summary - Section 2

Bernoulli's equation (potential flow):

$$p + \frac{1}{2}\rho U^2 = \text{constant}$$

C_p for potential flow:

$$C_p = \frac{p - p_\infty}{1/2\rho_\infty U_\infty^2} = 1 - \left(\frac{U}{U_\infty}\right)^2$$

Potential flow BC:

Far from the object: $\vec{V} = \vec{U}_\infty$

On the surface: $\vec{V} \cdot \hat{n} = 0$

Summary - Section 2

Uniform potential flow:

$$u = U_{\infty} \cos \alpha$$

$$v = U_{\infty} \sin \alpha$$

$$\phi = U_{\infty} (x \cos \alpha + y \sin \alpha)$$

$$\psi = U_{\infty} (y \cos \alpha - x \sin \alpha)$$

$$\phi = U_{\infty} r (\cos \theta \cos \alpha + \sin \theta \sin \alpha)$$

$$\psi = U_{\infty} r (\sin \theta \cos \alpha - \cos \theta \sin \alpha)$$

Source/sink potential flow:

$$v_r = \frac{K}{2\pi r}$$

$$v_{\theta} = 0$$

$$\phi = \frac{K}{2\pi} \ln r$$

$$\psi = \frac{K}{2\pi} \theta$$

$$K > 0 \text{ source}$$

$$K < 0 \text{ sink}$$

Summary - Section 2

Doublet potential flow:

$$v_r = -\frac{B}{2\pi} \frac{\cos\theta}{r^2}$$
$$\phi = \frac{B}{2\pi} \frac{\cos\theta}{r}$$

$$v_\theta = -\frac{B}{2\pi} \frac{\sin\theta}{r^2}$$
$$\psi = -\frac{B}{2\pi} \frac{\sin\theta}{r}$$

Vortex potential flow:

$$v_r = 0$$
$$\phi = -\frac{\Gamma}{2\pi} \theta$$

$$v_\theta = -\frac{\Gamma}{2\pi r}$$
$$\psi = \frac{\Gamma}{2\pi} \ln r$$

Summary - Section 2

Flow over a non-spinning cylinder:

$$\psi_{\text{non-spinning}} = \psi_{\text{uniform flow}} + \psi_{\text{doublet}} \rightarrow l = 0, \quad d = 0$$

Flow over a spinning cylinder:

$$\psi_{\text{spinning}} = \psi_{\text{uniform flow}} + \psi_{\text{doublet}} + \psi_{\text{vortex}} \rightarrow l = \rho_{\infty} U_{\infty} \Gamma, \quad d = 0$$

Summary - Section 3

Wall shear stress:

$$\tau_w = \mu \left[\frac{\partial u}{\partial y} \right]_{y=0}$$

Summary - Section 4

Kutta-Joukowski Theorem

$$l = \rho_{\infty} U_{\infty} \Gamma$$

Summary - Section 4

Thin airfoil theory

Fundamental eq. of thin airfoil theory:

$$U_{\infty} \left(\alpha - \frac{dz}{dx} \right) = \frac{1}{2\pi} \int_0^{\pi} \frac{\gamma(\theta) \sin \theta}{(\cos \theta - \cos \theta_0)} d\theta$$

$$x = \frac{c}{2} (1 - \cos \theta_0), \quad 0(\text{LE}) \leq \theta_0 \leq \pi(\text{TE})$$

$$\Gamma = \frac{c}{2} \int_0^{\pi} \gamma(\theta) \sin \theta \, d\theta$$

Symmetric airfoil:

$$\gamma(\theta) = 2\alpha U_{\infty} \frac{1 + \cos \theta}{\sin \theta}$$

$$C_{m0} = \frac{-C_l}{4}$$

Summary - Section 4

Thin airfoil theory (contd.)

Cambered airfoil:

$$\frac{dz}{dx} = (\alpha - A_0) + \sum_{n=1}^{\infty} A_n \cos(n\theta_o)$$

$$A_0 = \alpha - \frac{1}{\pi} \int_0^{\pi} \left(\frac{dz}{dx} \right) d\theta, \quad A_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{dz}{dx} \right) \cos(n\theta) d\theta$$

$$\gamma(\theta) = 2U_{\infty} \left[A_0 \frac{1 + \cos\theta}{\sin\theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right]$$

$$C_l = \pi (2A_0 + A_1)$$

$$C_{m0} = - \left[\frac{C_l}{4} + \frac{\pi}{4} (A_1 - A_2) \right]$$

$$\frac{x_{cp}}{c} = \frac{1}{4} \left[1 + \frac{\pi}{C_l} (A_1 - A_2) \right]$$

$$C_{mc/4} = \frac{\pi}{4} (A_2 - A_1)$$

Summary - Section 5

Horse shoe vortex

Velocity induced by vortex line extending to ∞ :

$$|\vec{V}| = \frac{\Gamma}{4\pi R}$$

Prandtl's lifting line theory

Fundamental Eqn. of Prandtl's lifting line theory:

$$\alpha(y_1) = \underbrace{\frac{\Gamma(y_1)}{\pi U_\infty c(y_1)} + \alpha_{0l}(y_1)}_{\alpha_e} + \underbrace{\frac{1}{4\pi U_\infty} \int_{-b/2}^{b/2} \left[\frac{d\Gamma}{dy} \right] \frac{dy}{(y_1 - y)}}_{\epsilon}$$

Once $\Gamma(y)$ is known:

1. $l(y) = \rho_\infty U_\infty \Gamma(y) \rightarrow$ Load distribution

Summary - Section 5

Prandtl's lifting line theory contd.

$$2. \quad L = \int_{-b/2}^{b/2} l(y) \, dy = \int_{-b/2}^{b/2} \rho_{\infty} U_{\infty} \Gamma(y) \, dy$$

$$C_L = \frac{L}{q_{\infty} S} = \frac{2}{U_{\infty} S} \int_{-b/2}^{b/2} \Gamma(y) \, dy$$

$$3. \quad d_v \approx l\epsilon \text{ (for small } \epsilon) \rightarrow d_v(y) \approx l(y)\epsilon(y)$$

$$= \rho_{\infty} U_{\infty} \Gamma(y) \left(\frac{-w(y)}{U_{\infty}} \right) = -\rho_{\infty} \Gamma(y) w(y)$$

$$D_v = \int_{-b/2}^{b/2} d_v \, dy = - \int_{-b/2}^{b/2} \rho_{\infty} \Gamma(y) w(y) \, dy$$

$$C_{Dv} = \frac{D_v}{q_{\infty} S} = \frac{2}{U_{\infty}^2 S} \int_{-b/2}^{b/2} \Gamma(y) w(y) \, dy$$

Summary - Section 5

Prandtl's lifting line theory contd.

Generalized lift distribution

$$\text{Let } y = -\frac{b}{2} \cos \theta = -s \cos \theta, \quad y_1 = -\frac{b}{2} \cos \theta_1$$

$$\text{and } \Gamma(y) = 2bU_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \rightarrow \text{Fourier sine series}$$

$$C_L = A_1 \pi \frac{b^2}{S} = A_1 \pi AR$$

$$C_{Dv} = \frac{C_L^2}{\pi e AR}, \quad e = \frac{1}{1 + \delta}, \quad \delta = \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \geq 0$$

Elliptical lift distribution

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b} \right)^2} \rightarrow A_1 = \frac{\Gamma_0}{2bU_\infty}, \rightarrow A_n = 0 \text{ (for } n \neq 1) \rightarrow \delta = 0 \text{ \& } e = 1$$